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SIMULATION OF TWO-PHASE FLOW AND HEAT TRANSFER IN MINI- AND MICRO-CHANNELS FOR CONCENTRATING PHOTOVOLTAICS COOLING

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ABSTRACT

Advances in concentrating photovoltaics technology have generated a need for more effective thermal management techniques. Research in photovoltaics has shown that there is a more than 50% decrease in PV cell efficiency when operating temperatures approach 60°C. It is estimated that a waste heat load in excess of 500 W/cm^2 will need to be dissipated at a solar concentration of 10,000 suns. Mini- and micro-scale heat exchangers provide the means for large heat transfer coefficients with single phase flow due to the inverse proportionality of Nusselt number with respect to the hydraulic diameter. For very high heat flux situations, single phase forced convection in micro-channels may not be sufficient and hence convective flow boiling in small scale heat exchangers has gained wider scrutiny due to the much higher achievable heat transfer coefficients due to latent heat of vaporization and convective boiling.

The purpose of this investigation is to explore a practical and accurate modeling approach for simulating multiphase flow and heat transfer in mini- and micro-channel heat exchangers. The work is specifically aimed at providing a modeling tool to assist in the design of a mini/micro-scale stacked heat exchanger to operate in the boiling regime. The flow side energy and momentum equations have been implemented using a one-dimensional homogeneous approach, with local heat transfer coefficients and friction factors supplied by literature correlations. The channel flow solver has been implemented in MATLAB™ and embedded within the COMSOL™ FEM solver which is used to model the solid side conduction problem. The COMSOL environment allows for parameterization of design variables leading to a fully customizable model of a two-phase heat exchanger.

INTRODUCTION

The recent trend towards two-phase mini- and micro-channel heat exchangers is fueled by the need for a compact heat sink capable of dissipating high heat fluxes. Micro-channel evaporators utilize the inverse proportionality of Nusselt number with respect to hydraulic diameter and the latent heat of vaporization to dissipate heat fluxes on the order of hundreds of watts per square centimeter. The large heat loads produced by concentrated solar power (CSP) make it a well-suited application for two-phase micro-channel heat sinks. Figure 1 shows a schematic of how a micro-channel heat exchanger could be implemented for use with concentrated photovoltaics (CPV).

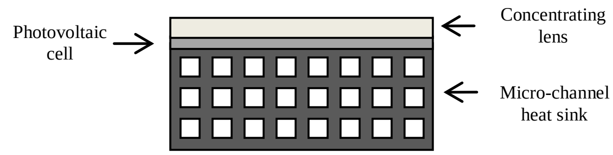

Figure 1 - Schematic of implementation of micro-channel heat sink on CPV

The highest operating efficiencies for multi-junction CPVs are currently about 41% [1]. This efficiency is improved with the use ultra-high concentrations of 1000 suns and greater. While the efficiency is predicted to increase linearly with increasing concentrations, it becomes more difficult to effectively cool UHCPV cells at such high concentrations [2]. The maximum solar radiation from a single sun is approximately 0.1 W/cm^2. This translates to a heat load of 100 W/cm^2 at a concentration of 1000 suns. Current two-phase micro-channel heat sinks are capable of dissipating heat loads of 300 W/cm^2 [3], with many researchers looking to increase the capability in excess of 500 W/cm^2.

The majority of the research in the area of boiling in micro-channels has focused on flow instabilities, flow mapping, local heat transfer coefficients, pressure drop and critical heat flux. While there is still much need for advances in these areas, there is an equal need for methods and tools for enabling practical design of two-phase heat sinks. Zhang et al. [4] use thermal resistance circuits to define the solid coupled to a finite volume approach to model the fluid, while Qu et al. [5] assume the channel walls act as a fin and implement a fin analysis with a streamwise varying heat transfer coefficient. While the fin analysis has been adopted by many investigators because of its ease of implementation for high aspect ratio channel, single layer designs, it lacks the versatility and customization necessary for practical design of more complex micro-heat exchanger configurations.

MOTIVATION AND OBJECTIVES

Currently, there are commercially available CPV systems that operate at concentrations of 500 suns. Cell costs in such systems represent 40% of total system costs, and the overall operation costs are $1.50-$2.00 per peak watt. If concentrations of 5000-9000 can be achieved by implementing proper cooling techniques, cell costs could be as low as 10% of total system costs with operation costs around $1.00 per peak watt [2].

Although micro-channel heat exchangers are an obviously good match for high heat flux CPV cooling, operation in the single phase regime will not provide sufficiently high thermal load management for concentrations from 1,000 to 10,000 suns, leading to heat fluxes from 100 to 1000 W/cm^2. Along with spray cooling, mini- and micro-channel heat exchangers operating in a two-phase convective boiling regime are the most promising thermal management technologies for these heat flux regimes. Most of the important developments in small scale heat exchangers operating in the boiling regime have been driven by micro-processor cooling applications; the great majority of the work has focused on simple heat exchanger designs consisting of single layers of rectangular channels in a semiconductor or metallic base material. In previous work, [6], the author's research group has extensively studied the behavior of stacked mini- and micro-channel heat exchangers consisting of many layers of low aspect ratio channels in ceramic or metallic base materials operating in the single phase regime. This class of heat exchangers has shown very high thermal performance at reduced pressure drop, compared to traditional single layer, high aspect ratio designs.

The objectives of the present paper are to extend the analysis of the stacked mini-channel heat exchangers into the convective boiling regime necessary for CPV thermal management. A secondary but equally important objective is to discuss a computational approach that was developed to enable the application of best available convective boiling correlations for heat transfer and pressure drop, within the framework of a general FEM solver for the heat exchanger analysis. An accurate model for predicting cell temperatures at concentrations that are not yet achievable would help facilitate the design of a proper cooling system to meet the demands of advanced CPV systems. The guidance of a computational model would allow for experimentation of complex geometries and materials that would accommodate the high heat flux demands of high concentration systems.

MODEL OVERVIEW

This section will describe the detailed development of a generalized model that was used to analyze the behavior of stacked mini-channel heat exchangers in a convective boiling regime. Figure 2 shows the geometry of the stacked micro-channel heat exchangers examined in this study.

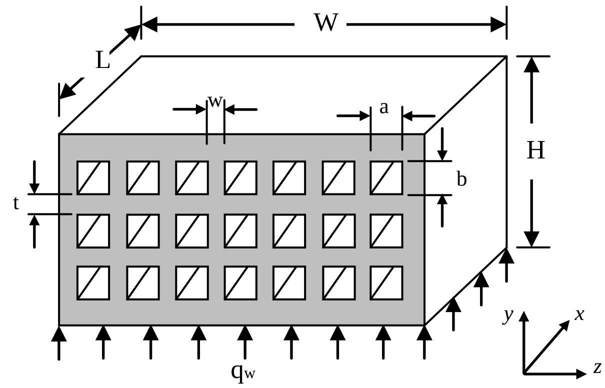

Figure 2 - Depiction of a multi-layer mini-channel heat sink with heat applied at the lower surface

The model utilizes a finite difference formulation of the governing equations with external call to computational state equations available in NIST Refprop [7]. The channel side energy and momentum equations are solved in a MATLAB™ framework. The MATLAB solver is embedded in a commercial FEM solver, COMSOL™, to enable a conjugate simulation of two-phase flow in a complex micro-channel heat exchanger that would be impossible to model using the simplified ad hoc procedures previously adopted for simple designs. The use of a flexible, commercial FEM tool allows for the study of complex geometries and materials and the parameterization of design variables. The end result is a fully customizable model of a two-phase micro-channel heat exchanger.

A one-dimensional, correlation based model is used to represent the boiling flow within channels. This model employs a simplified, pseudo-fluid approach that eliminates the need to track fluid-vapor interfaces. The effect of the fluid-vapor interface on heat transfer is not yet accurately predicted with current correlations, and thus this information would not augment the accuracy of any practical formulation.

Equations 1-4 represent standard formulations of mass, energy, and momentum conservation equations assuming one-dimensional, two-phase flow [8]. The equations were discretized using explicit finite difference formulations. Since the approach is one-dimensional, z represents the streamwise direction. The present model neglects the effects of kinetic and potential energy, shown by the crossed out terms in Eq. 2 and 4.

Mass:

$$\rho_m UA = const \tag{1}$$

Momentum:

$$\dot{m}\frac{du}{dz} = -A\frac{dp}{dz} - P\tau_w - \cancel{A\rho_m \cos\theta} \tag{2}$$

or rearranging for $\left(\frac{dp}{dz}\right)$

$$\frac{dp}{dz} = -\frac{\dot{m}}{A}\frac{du}{dz} - \left(\frac{P}{A}\right)\tau_w \tag{3}$$

Energy:

$$q''_{eff} P = \dot{m}\frac{d}{dz}\left(h + \cancel{v^2/2} + \cancel{gy}\right) \tag{4}$$

For the single-phase regime, Eq. 3 takes on the form of Eq. 5, where f_{eff} is the effective Fanning friction factor accounting for both frictional and accelerational pressure drop.

$$\left(\frac{dp}{dz}\right)_{sp} = \frac{2 f_{eff} \rho u^2}{D_h} \tag{5}$$

For the two-phase regime, Eq. 3 takes on the form of Eq. 6, where $\left(\frac{dp}{dz}\right)_{tp,f}$ is the frictional pressure gradient and $\left(\frac{dp}{dz}\right)_{tp,a}$ is the accelerational pressure gradient.

$$\left(\frac{dp}{dz}\right)_{tp} = \left(\frac{dp}{dz}\right)_{tp,f} + \left(\frac{dp}{dz}\right)_{tp,a} \tag{6}$$

Here $\left(\frac{dp}{dz}\right)_{tp,f}$ is defined by Kanlikar et al. [9] by Eq. 7, where ϕ_L is the two-phase multiplier defined by Eq. 8.

$$\left(\frac{dp}{dz}\right)_{tp,f} = \left(\frac{dp}{dz}\right)_{sp} \phi_L^2 \tag{7}$$

$$\phi_L^2 = 1 + \frac{C}{X} + \frac{1}{X^2} \tag{8}$$

X is the Martinelli parameter, which is defined as the ratio of the frictional pressure drop associated with a liquid only flow to the frictional pressure drop associated with a vapor only flow, as shown by Eq. 9.

$$X^2 = \left(\frac{dp}{dz}\right)_{f,LO} \Big/ \left(\frac{dp}{dz}\right)_{f,VO} \tag{9}$$

C is a constant that is empirically derived. The present model uses the value of C characterized by Mishima and Hibiki [10] by Eq. 10, where D_h is the hydraulic diameter of the channel in meters.

$$C = 21\left(1 - e^{-319 D_h}\right) \tag{10}$$

The pressure drop associated with the acceleration of the fluid, Eq. 11, is derived from the presence of bubbles, or voids, in the fluid flow. The bubbles form in cavities on the surface of the channel and grow until the buoyancy forces are great enough to lift it from the surface. The fluid then accelerates to fill in the now vacant cavity on the surface.

$$\left(\frac{dp}{dz}\right)_{tp,a} = G^2 v_{LV} x \tag{11}$$

The mixed mean temperature of the fluid is determined by the relationship to the heat transfer coefficient as defined by Eq. 12.

$$q''_{eff} = H\left(T_w - T_m\right) \tag{12}$$

For the single-phase regime, the heat transfer coefficient is determined by the relationship to Nusselt number, shown in Eq. 13, where Nusselt number values are interpolated from a table given by Kanlikar et al. for simultaneously developing flow in [9],

$$Nu = \frac{H_{LO} D_h}{k_l} \tag{13}$$

where H_{LO} is the liquid only or single phase heat transfer coefficient. The two-phase heat transfer coefficient is determined by empirically based correlations found in the literature. Kandlikar et al. [9] present a novel correlation that accounts for the heat transfer coefficient associated with nucleate boiling dominated flow and convective boiling dominated flow. This correlation is implemented in the present model due to its ability to characterize both the sub-cooled and saturated boiling regimes, and is defined by Eq. 14, 15, and 16.

$$h_{tp,NBD} = 0.6683 Co^{-0.2} (1-x)^{0.8} h_{LO} + 1058.0 Bo^{0.7} (1-x)^{0.8} F_{Fl} h_{LO} \quad (14)$$

$$h_{tp,CBD} = 1.136 Co^{-0.9} (1-x)^{0.8} h_{LO} + 667.2 Bo^{0.7} (1-x)^{0.8} F_{Fl} h_{LO} \quad (15)$$

$$h_{tp} = \text{larger of} \begin{cases} h_{tp,NBD} \\ h_{p,CBD} \end{cases} \quad (16)$$

Bo, the boiling number, is defined by Eq. 17 and Co, the convection number, is defined by Eq. 18.

$$Bo = \frac{q''_{eff}}{G h_{LV}} \quad (17)$$

$$Co = \left[(1-x)/x\right]^{0.8} \left(\rho_V \big/ \rho_L \right)^{0.5} \quad (18)$$

F_{Fl} is the fluid-surface parameter used to account for the influence of different fluids on a brass or copper surface, as defined by Kandlikar et al. in [9].

MODEL ALGORITHM

The solution of the model relies on the convergence of two parameters; the pressure drop in each channel and the heat load on each channel. The problem is initiated with a guess of the distribution of heat flux and pressure drop for each channel. The inlet fluid temperature, inlet fluid pressure, and header inlet mass flow rate are required input. The problem is iterated in MATLAB until the local heat transfer coefficient distribution and local mixed mean fluid temperature for each channel is determined. These distributions are then imported into COMSOL as convective cooling boundary conditions for each channel and the finite element model is solved. An average heat flux for each channel is extracted from the COMSOL solution and applied as the new heat load for the MATLAB solver. This process is repeated until the heat flux in each channel no longer changes. Once the solution is converged on heat flux, the pressure drop in each channel is compared. If the pressure drop from one channel to the next differs, the mass flow rate in each channel is adjusted and the model is allowed to converge on heat flux again. This process is repeated until the pressure drop in each channel is the same. Figure 3 shows a schematic of this algorithm.

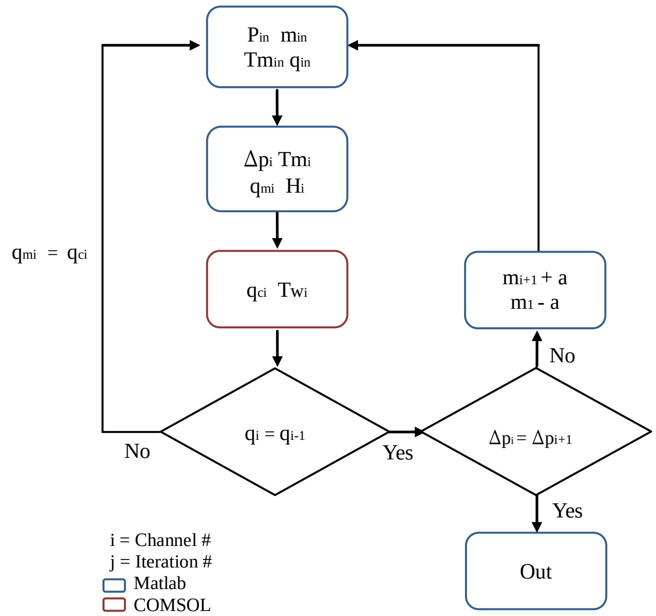

Figure 3 - Schematic of solution algorithm

COMPARISON WITH EXPERIMENTAL DATA

The present model was compared to experiments conducted by Qu and Mudawar [5]. The experiments were performed using a copper heat sink with 21 channels measuring 231µm in width by 713µm deep. The channels were 4.48 cm in length with a channel wall thickness of 118µm. The channels were heated from below, through the 2462µm copper base. The channels were covered with an insulating plastic cover plate measuring 12700µm thick. One channel, with the given dimensions and materials was simulated with the model for comparison.

The experiments were run for a heat flux range of 40 – 130 W/cm^2 and varying mass fluxes. The comparison was conducted using a mass flux of 255 kg/m^{2}s. Qu et al. compared their experimental results with correlations from the literature and found that a correlation by Yu et al. [11] fit the data for heat transfer coefficient best and the correlation by Mishima and Hibiki [10] fit the data for pressure drop. They used a mean absolute error equation, Eq. 19, to determine the correlation with the best fit to the data.

$$MAE = \frac{1}{M} \sum \frac{\left|\Delta P_{pred} - \Delta P_{\exp}\right|}{\Delta P_{\exp}} \times 100 \quad (19)$$

The present model was implemented with the correlations outlined by Qu et al. as fitting their data best. Figure 4 shows a comparison of the present model with the local heat transfer coefficient determined by the experiment and the correlation as plotted by the investigators.

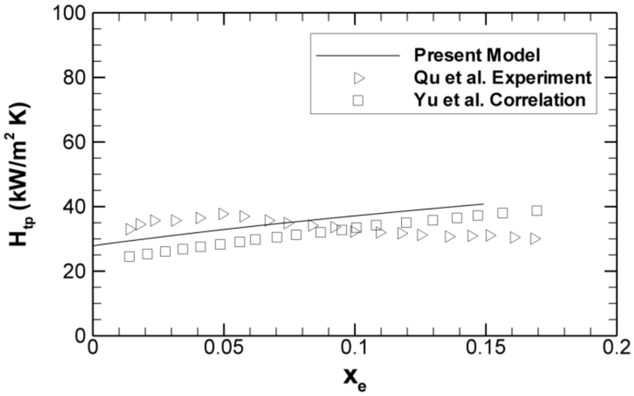

Figure 4 – Local heat transfer coefficient comparison with experimental results

The correlation by Yu et al. was determined to have a MAE of 17.57%, while the present model fit the data to within 17.77% using the same correlation. This shows that the present model predicts the local heat transfer coefficient as accurately as the correlation that is used.

Figure 5 shows the results of pressure drop for the range of heat fluxes investigated.

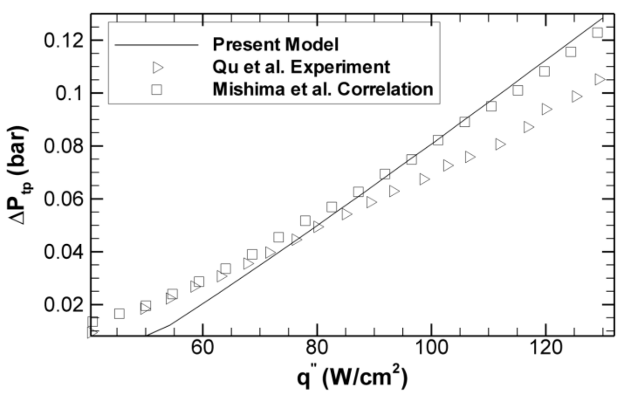

Figure 5 - Pressure drop comparison with experimental data

The model is consistent with the correlation by Mishima et al. for a large range of heat fluxes. It should be noted that the experiments were conducted with a fixed outlet pressure, while the simulation is conducted with a fixed inlet pressure. This discrepancy creates a shift in the data predicted by the model. Despite the known source of error, the model predicts the pressure drop to within 17.30%, while the correlation was determined to fit the data to within 12.23%. The poor agreement with the experimental data and the correlation at low heat fluxes could be explained by the model's criteria for transition to the boiling regime. Figure 4 shows the model under predicting the heat transfer coefficient at low heat fluxes, suggesting that the model has not transitioned to saturated boiling flow which would serve to increase the heat transfer coefficient. The under prediction of the transition to saturated boiling flow would also serve to lower the predicted pressure drop by the model, as seen in Figure 5.

STACKED MULTI-LAYER MODEL RESULTS

The stacked multi-layer mini-channel heat exchanger shown in Fig. 2 was modeled using the computational model described here. A six layer design was chosen, as shown in Figure 6. Because the channels are evenly spaced with the same configuration in each layer, it was possible to computationally model only a single column of channels, with the symmetry surfaces on both sides modeled as adiabatic boundary conditions. The streamwise length of the heat exchanger, L, was arbitrarily chosen as 25mm because this represents the length of our laboratory test devices. Low and a high heat flux cases were studied, using 50 W/cm^2 and 200 W/cm^2, respectively. The heat flux was assumed to be applied uniformly to the lower surface of the heat exchanger, as seen in Fig. 2. Both cases use R245fa as the working fluid, and copper as the heat sink material. The geometric dimensions and inlet conditions for each case are outlined in Table 1.

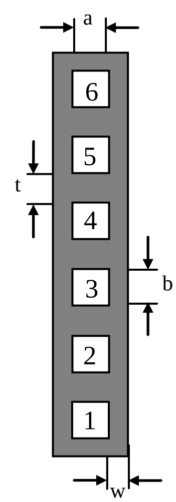

Figure 6 - Six channel geometry dimensions

	50 W/cm^2	200 W/cm^2
Fluid Temperature	12°C	2°C
Pressure	600 kPa	600 kPa
Mass flow rate	0.3 g/s	0.3 g/s
a	200 µm	200 µm
b	200 µm	200 µm
t	250 µm	250 µm
w	125 µm	125 µm

Table 1 – Model parameters

LOW HEAT FLUX RESULTS

As expected, the surface temperature distribution for the heat sink, shown in Figure 7, shows an increase in temperature from inlet to outlet, and a decrease in temperature moving away from the heat source. It is interesting to note that the isotherms have a diagonal orientation suggesting a decrease in heat flux the further the channel is from the surface where heat is applied, in this case, the lower surface.

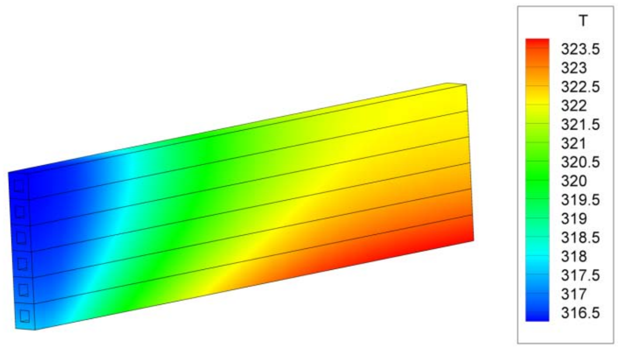

Figure 7 – Low heat flux heat sink surface temperature distribution

This trend can be explained by analyzing the heat load on each channel. Figure 8 shows the ratio of channel heat load to the total heat load for the heat sink for each channel. It can be seen that the heat load decreases moving away from the heat source, and in fact approaches a constant ratio. The asymptotic trend is surprising in that the heat flux to the upper channels does not vanish. This is very different from the behavior of the equivalent heat exchanger in single phase flow.

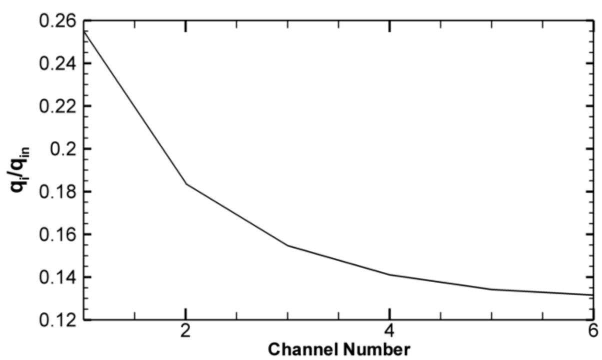

Figure 8 – Low heat flux heat load ratio

This notion is further confirmed by analyzing the local thermodynamic quality of the fluid in each channel. As seen in Figure 9, the quality follows a linearly increasing trend as a function of streamwise position in the channel. The quality in channels 5 and 6 are virtually identical and barely exceed zero, signifying minimal phase change along the length of the channel. Because channels 5 and 6 are in single phase flow, the flow rate to each of them is the same, and so too is the heat transfer coefficient and therefore the absorbed heat flux. It is readily apparent that the upper channels underperform because they do not enter into the convective boiling regime.

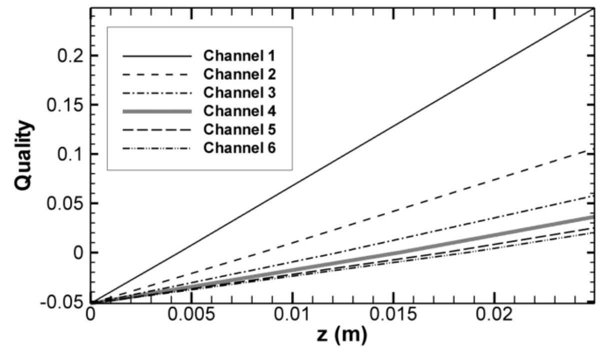

Figure 9 – Low heat flux local quality distribution

The virtually single-phase nature of the top two channels causes flow to be diverted away from the channels sustaining the highest heat loads. This is due to the relatively low pressure drop incurred by single-phase flow as compared to saturated boiling. Figure 10 shows the ratio of mass flow rate in each channel to the total mass flow rate. It can be seen from Figures 8 and 10 that the top two channels are supplied with over 40% of the total mass flow, while sustaining less than 30% of the total heat load.

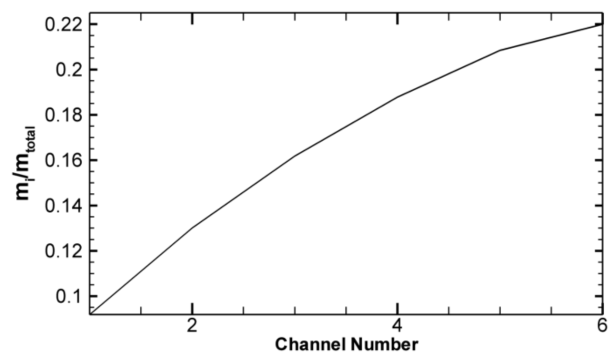

Figure 10 - Low heat flux mass flow rate ratio

While the top two channels thermally underperform, they do serve to decrease the overall pressure drop of the heat sink. Figure 11 depicts the local pressure of the fluid in each channel.

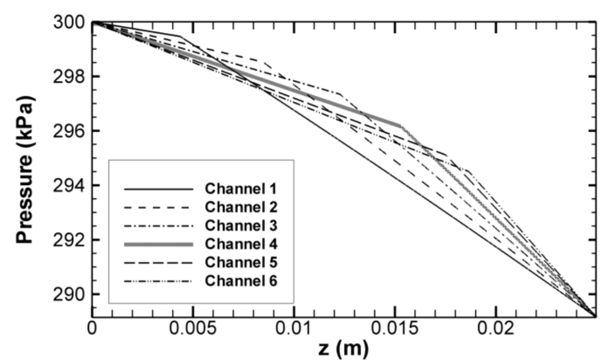

Figure 11 – Low heat flux local pressure for each channel

The inflection point in each curve signifies a transition to saturated boiling and is generated by the large pressure drop incurred by boiling flow. It is worth noting that for relatively low heat loads, the transition to saturated boiling occurs at significantly different streamwise locations for each channel. This is important because it would suggest that fewer channels would allow for better utilization of the latent heat of the working fluid, and thus multi-channel heat sink designs may not be necessary for low heat flux situations.

HIGH HEAT FLUX RESULTS

Diagonal isotherms can again be seen in the surface temperature distribution for the larger heat load, as shown in Figure 12.

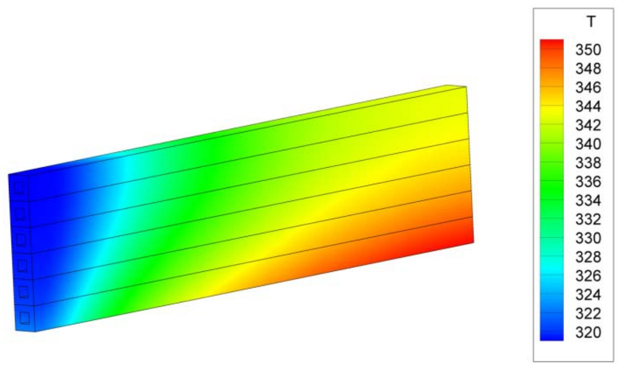

Figure 12 - High heat flux heat sink surface temperature distribution

The shallow slope of the isotherms towards the outlet of the channel suggests a larger degradation of heat load than in the low heat flux case. A plot of the heat load ratio for each channel, shown in Figure 13, supports the notion.

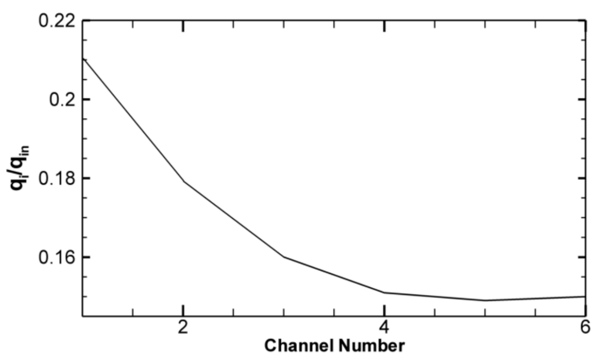

Figure 13 - High heat flux heat load ratio

Again, the heat load ratio approaches a constant value, but suggests that for the higher heat load only the top channel is providing little thermal enhancement. The local thermodynamic quality for each channel at higher heat loads is shown in Figure 14. It can be seen that all of the channels have reached saturated boiling, with the top two channels only differing slightly from one another.

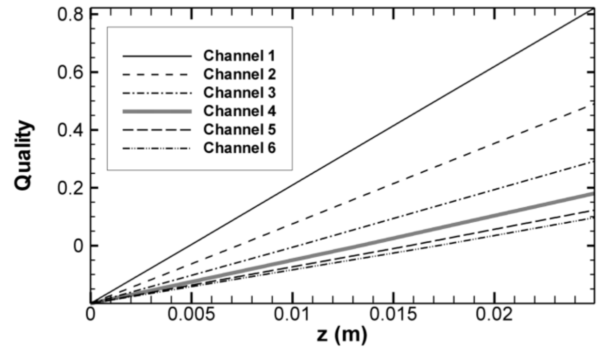

Figure 14 - High heat flux local quality distribution

It is important to note that the magnitudes of the quality for the higher heating case are much higher than those of the low heat flux case. This is evidence to the fact that the latent heat of the working fluid is better utilized at higher heat loads. However, we again see that the channels absorbing the least amount of heat are supplied with the most amount of flow, as shown in Figure 15. It is worth noting that for the higher heating case the top three channels account for over 65% of the total mass flow, but are only absorbing about 45% of the total heat load. This is due to the fact that the two-phase pressure drop is a strong function of thermodynamic quality. As seen in Figure 14, there is a large change in quality from channel to channel, thus causing the pressures in each channel to vary significantly. This causes the flow to divert to the channels with the lowest quality, and therefore the lowest pressures.

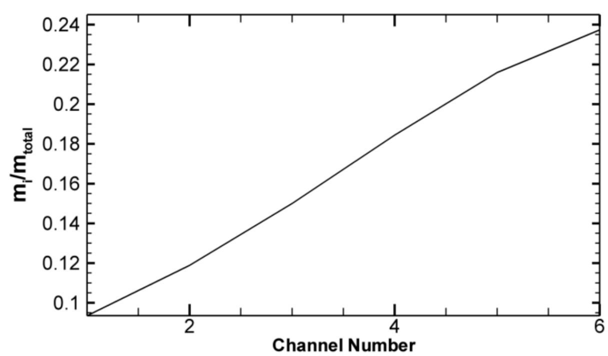

Figure 15- High heat flux mass flow rate ratio

The local pressure distribution for each channel is similar to those of the lower heating case. Figure 16 shows the transition to saturated boiling occurring earlier for each channel, which is consistent with Figure 14.

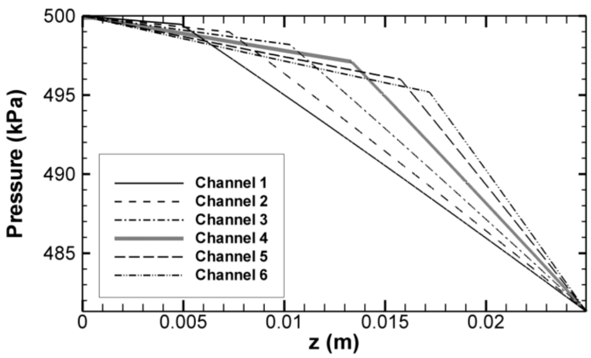

Figure 16 - High heat flux local pressure distribution

The overall pressure drop for the higher heating case is much higher than that of the low heat flux case. This is again due to the higher qualities that are present at higher heat fluxes. The larger qualities increase the effect of the accelerational pressure drop.

CONCLUSIONS

The present model displays the ability to capture conjugate effects in two-phase flow in micro-channels. The model exhibits trends consistent with multi-phase flow and conduction in a solid. The model was compared with experiments performed on copper micro-channels with an insulating plastic lid and was determined to fit the data with an acceptable degree of accuracy. The model was exercised using low and high heat loads and provided insight into the optimal number of channels needed to meet the thermal demands of the system. The pressure drop for the different heating cases was analyzed and shown to be significantly higher for higher heat loads, which is consistent with what would be expected from higher qualities in the channels.

NOMENCLATURE

A	channel cross-sectional area
Bo	boiling number
C	Martinelli-Chisholm constant
Co	convection number
D_h	hydraulic diameter
F_{FL}	fluid-surface parameter
G	mass flux
g	gravitational constant
H_{LO}	liquid-only heat transfer coefficient
H_{tp}	two-phase heat transfer coefficient
$H_{tp,CBD}$	convective boiling dominated heat transfer coefficient
$H_{tp,NBD}$	nucleate boiling dominated heat transfer coefficient
h	enthalpy
h_{LV}	latent heat of vaporization
M	number of data points
$\dot{m}$	mass flow rate
P	Perimeter
ΔP_{tp}	two-phase pressure drop
$\Delta P_{tp,a}$	two-phase accelerational pressure drop
$\Delta P_{tp,f}$	two-phase frictional pressure drop
p	pressure
q''_{eff}	effective heat flux
u	fluid velocity
v	kinetic energy velocity
v_{LV}	specific volume difference between saturated vapor and saturated liquid
X	Martinelli parameter
x	quality
y	potential energy height
z	streamwise position

Greek symbols

ρ	density
τ_w	wall shear stress
θ	angular position
ϕ^2_L	two-phase frictional multiplier

Subscripts

L	liquid
V	vapor
LO	liquid-only
VO	vapor-only
tp	two-phase
sp	single-phase
f	frictional
a	accelerational
m	mixed-mean
w	wall
pred	predicted
exp	experimental

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